B. Tech. (Computer Engineering) Vth Semester

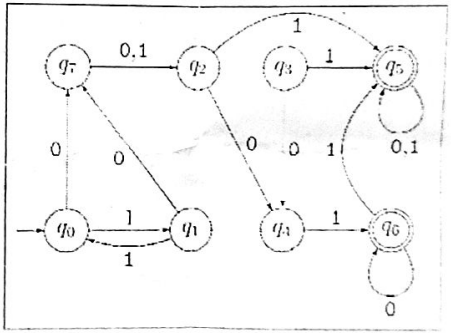
1st Sessional 2022-23, AUTOMATA THEORY (CEN-502)

Time: 1 Hour | MM:15

Note: Attempt all questions. | Date: 21/09/22

Your Roll No : 20BCS001

Q No.	CO Mapped
1	1
2	1
3	2
4	2

Q. No	Question	Mark
1.	Design a DFA for the language L_1 ($\sum$ = {0, 1}), such that each string w in the Langauge has at least as many occurrences of (110)'s as (011)'s. ***Verify your answer using suitable examples.***	3
2.	Minimize the following DFA using the Matrix Method.  ***Verify your answer using suitable examples.***	3
3	Design a DFA M, and also its corresponding Regular Expression R over $\sum$= {a, b, c} such that the language determined by R consists of all strings that contain at least one occurrence of each symbol in $\sum$. ***Explain your answers using suitable examples.***	2+3
4	Using Pumping Lemma, comment on the Regularity/ Non Regularity of the following languages : i) L_2 = { w , such that $\|w\|$ is prime , where $w \in \{0, 1\}^*$ } ii) L_3 = { w , such that #0 (w) != #1 (w) , where $w \in \{0, 1\}^*$}	4